University of Rwanda
College of Science and Technology
School of ICT
Department of Computer and Software Engineering

Year 3

Module Title : Mobile Application Systems and Design

Reg No : 223007592

Date : 17th March, 2026.

Flutter Lab 4 : Consuming APIs in Flutter.

Qn 1. In this project, I used the following dependencies;

- Provider : This is used for state management and helps to manage and share data (posts) accross the application efficiently.
- HTTP : This is used to make API requests such as GET, POST, PUT, and DELETE to communicate with the JSONplaceholder API.
- dart : convert : This is used to convert JSON data into Dart objects and vice versa.
- flutter/material.dart : This is used to design the user interface using Material Design components.

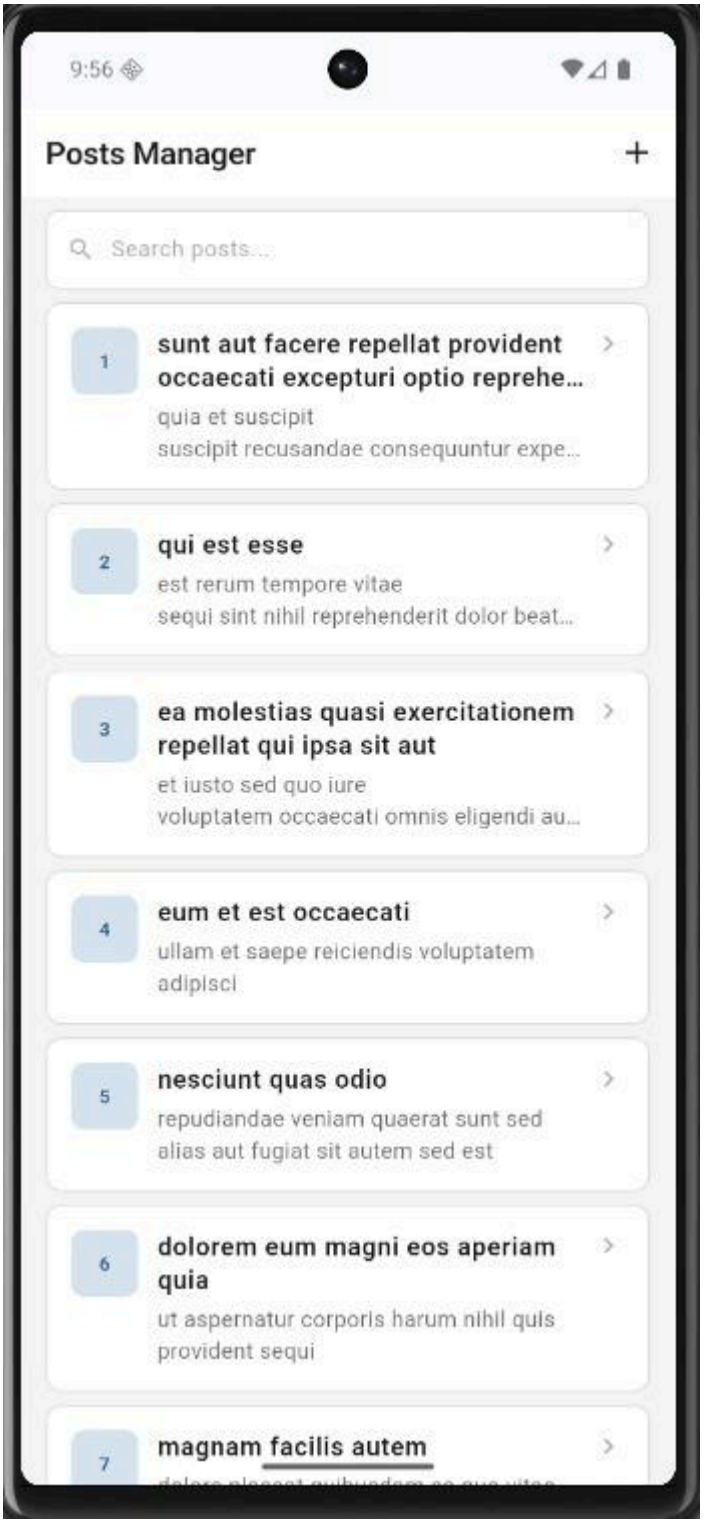
These dependencies help the application manage state, communicate with the API, process data, and display it properly in the UI.

Qn 2.
- Try-catch block : API calls are wrapped in, to handle errors like network failure.
- checking response status codes : The response from the API is checked (Eg status code 200 for success). If it fails, an error is handled.
- Handling network issues : For instance, if there is no internet connection, the app can display a message to the user.
- Validating data : The app ensures that the received data is valid before using it.
- Using provider for error state : Errors can be stored in the provider and ~~stored~~ displayed in the UI.

Qn 3. A Future is used to fetch data from the API asynchronously. In this project, Provider handles the Future and updates the UI using notifyListeners(). The post list depends on this data and rebuilds automatically when it is loaded.

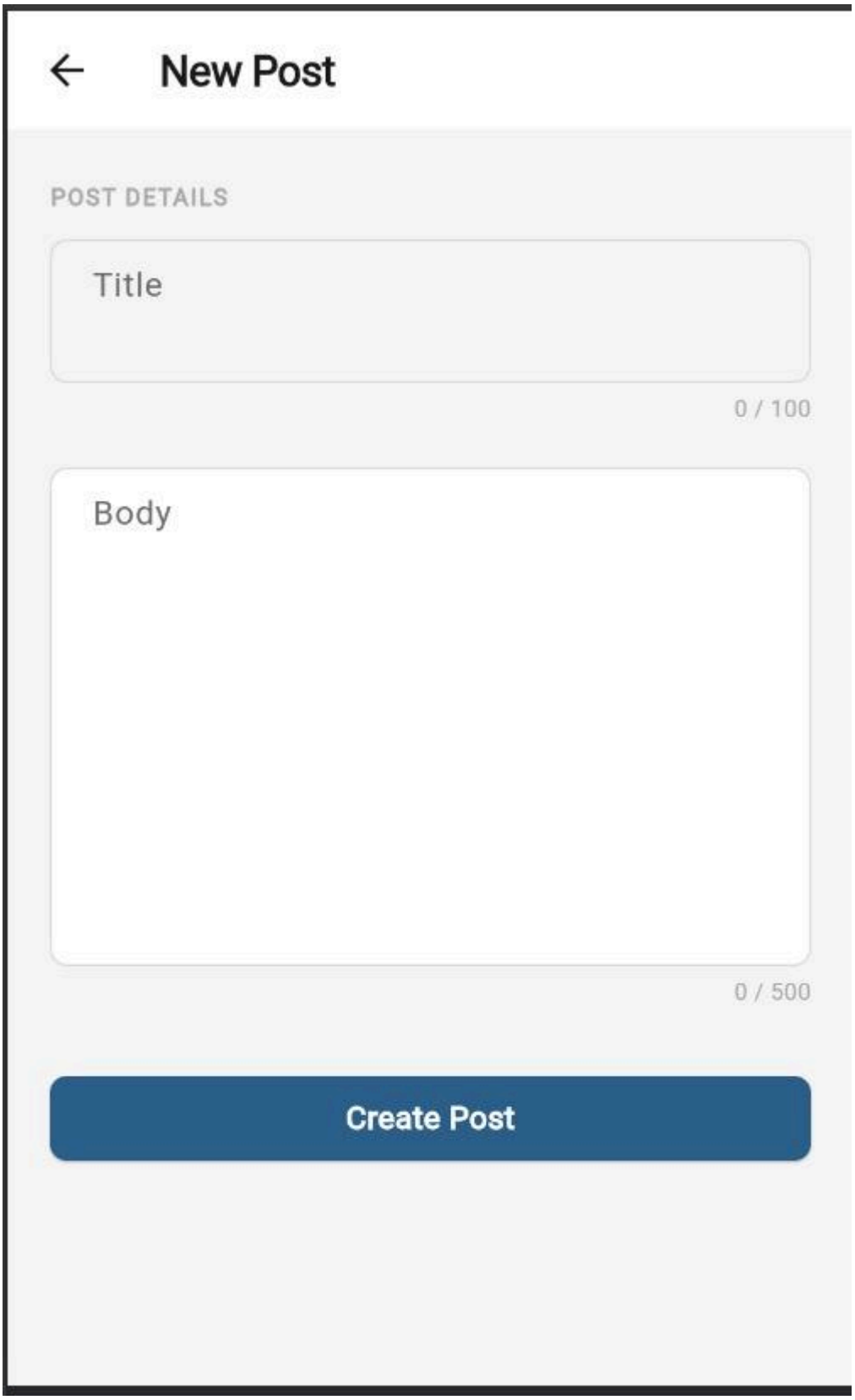
SCREENSHOTS.

I. Viewing all available posts

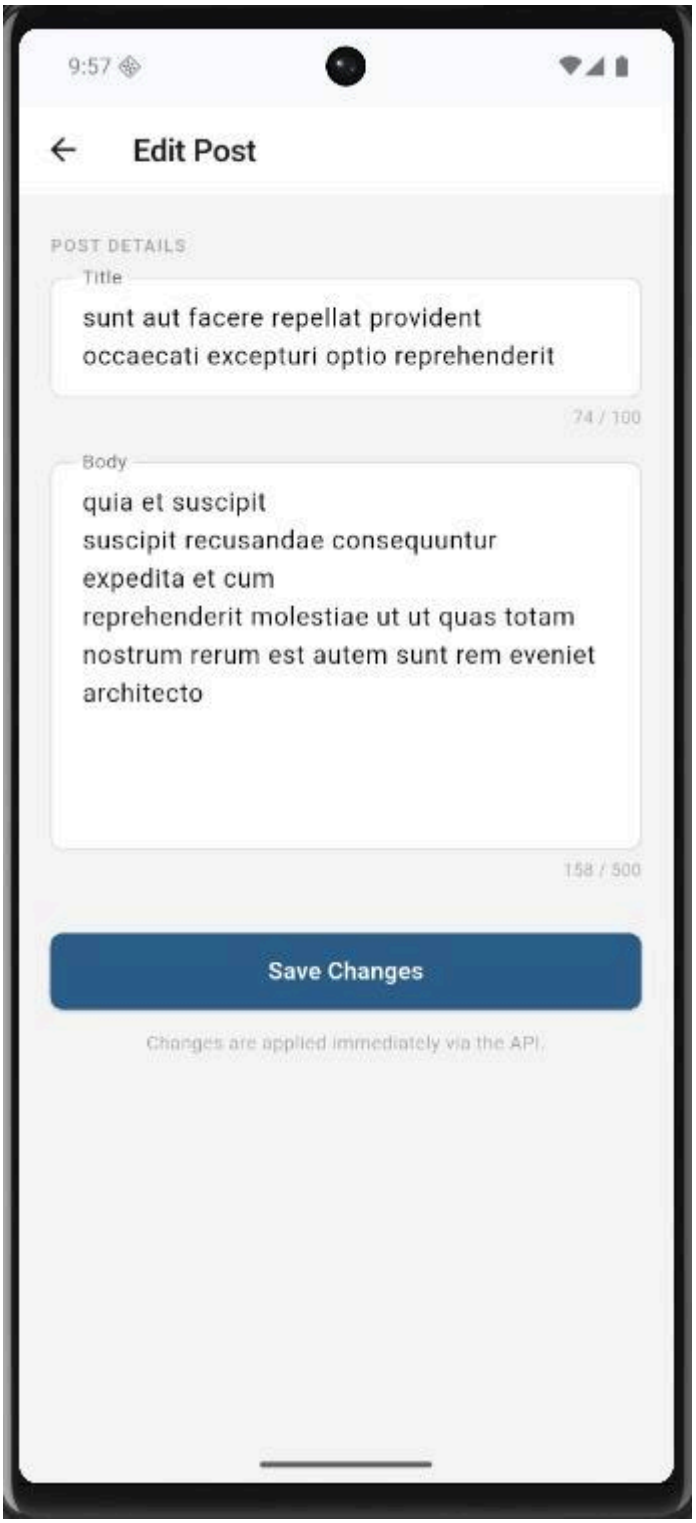
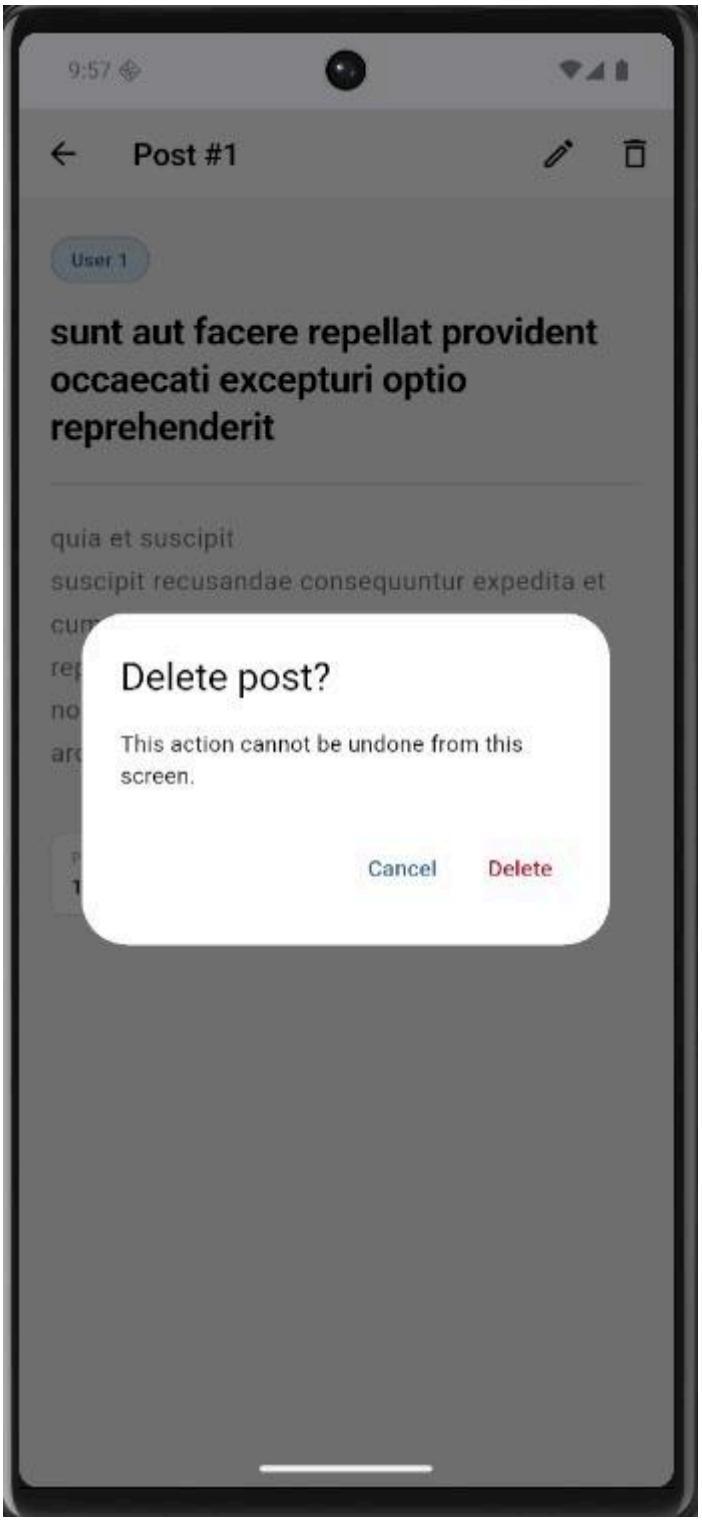


II. Reading details of a post

III. Creating a new post



IV. Editing an existing post



V. Delete a post